

1. Purpose

To define the process to add Administration on Behalf Of (AOBO) to a subscription.

AOBO allows TD Synnex to support the end user tenant and allows the reseller to earn a discount off of list (PEC).

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3. Procedure

3.1. AOBO

3.1.1. Why Does TD SYNnex Need AOBO

[Administration on Behalf Of](#) (AOBO), sometimes referred to as foreign principal, is related to access to an Azure subscription. AOBO allows members of the provider's/reseller's AdminAgents security group to access Azure subscriptions with the roles assigned.

AOBO is automatically provided to the provider when a new subscription is created. If removed or if a subscription is transferred from another provider, then AOBO must be added using a PowerShell script.

Note that the provider and reseller do not need any RBAC roles to sell services to the end user.

In the CSP program the indirect reseller gets a discount for supporting the end user customer. This discount is in the form of a credit called [Partner Earned Credit](#) (PEC). Microsoft confirms that the reseller is eligible for the credit by checking for a minimum level of access provided to the provider.

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If this access is removed, then the cost for the services billed to the reseller will increase to slightly more than MSRP until 24 hours after access is restored.

Although the permissions affect many areas the most obvious effects are:

1. No support tickets for Azure issues can be raised;
2. Azure Reserved Instances and Savings Plans cannot be purchased; and,
3. Access to subscriptions cannot be provided.

The resellers discount and ability to open support cases can be resolved by giving the provider the [RBAC](#) role "Support Request Contributor". This should be considered the least privilege that the provider would need.

In order to sell Reserved Instances or Savings Plans the provider would need the RBAC role

It is important to note that these permissions interact with the Granular Delegated Administration Permissions. For example, without AOBO Owner and GDAP Directory Readers access to reservations and subscription cannot be given. The Directory Readers Azure AD role allows the listing of user ids which may be given access.

3.1.2. Script Components

The PowerShell script is made up of cmdlet and some parameters. The script used to add AOBO is made up of the following components:

New-AzRoleAssignment	This is the cmdlet creates a new role assignment for a user, group, service principal, or managed identity in Azure.
-RoleDefinitionName "<ROLE_NAME>"	This parameter specifies the name of the role to assign. It may be any of the RBAC roles available in Azure and is the name of the role which is case sensitive.
-ObjectId "<ADMINAGT_OBJECT_ID>"	This parameter specifies the ID of the user, group, service principal, or managed identity to assign the role to. For AOBO, it will be the object ID of the AdminAgents group at TD SYNEX.
-ObjectType "ForeignGroup"	This parameter specifies the type of object being assigned the role. In this case, it's a "ForeignGroup", which typically refers to a group from a different Azure AD tenant.
-Scope "/subscriptions/<SUB_ID>"	This parameter defines the scope of the role assignment. It could be a subscription, a resource group, or a specific resource. In this case, it's a subscription, and you would replace <subscription_id> with the actual ID of the subscription where you want to assign the role.

3.2. TD Synnex AdminAgent Groups Object IDs

UNITED STATES

PAC	Object ID
Stellr	b13ae2dd-2d47-4211-89de-5142f8368588
SIE	aa7face0-42ae-47f1-a752-776945feb7fc

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SCM	9967c88e-30d5-476f-ab06-0b849ba52bde
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PAC	Object ID
Legacy Synnex	acd31259-050b-4f6d-a1ba-0fe854bb0181
Legacy TechData	28965048-c414-4fdb-b297-6e1c0d373331

3.3. Role Definition Names

The role definition names are the names of RBAC roles in Azure. Although any role would work with the script, not all will function as intended. The two recommended roles are listed below, should you wish to use a different role, please check with TDS engineering first.

Owner – this is the recommended role and the only one that works with legacy Tech Data platforms. It provides full control of the subscription.

Support Request Contributor – this role only allows TDS to open Azure support requests.

Note: Without the owner role, TDS will not be able to assign permissions, will be unable to provision reservations / savings plans, or run Click2Run solutions.

Note: The reseller will not get a discount (Partner Earned Credit) on a subscription unless the appropriate permissions are applied.

An additional role that may be added using AOBO is:

Reservation Purchaser – this role allows TDS to purchase/exchange/cancel reservations. It does not provide PEC; therefore, it would have to be added in addition to a role that supports PEC.

3.4. Errors

Some potential error messages you may encounter are:

Bad Request –

- Something about the request is malformed. Check spacing and characters.
- A foreign login is trying to run the command
- TD SYNEX does not have a reseller relationship with the end user tenant
- The subscription ID or object ID is wrong

Conflict – the role requested already exists in the subscription.

Forbidden – the user does not have the appropriate permission to add the foreign principal to the subscription. This means they are not a local account that has owner permissions.

3.4.1. Troubleshooting

If troubleshooting assistance from Microsoft is required, we will need the newRoleAssignment.log file.

Obtain the log file by appending the following to the script:

```
-Debug > newRoleAssignment.log
```

For example:

```
New-AzRoleAssignment -RoleDefinitionName "<ROLE_NAME>" -ObjectId  
"<ADMINAGT_OBJECT_ID>" -ObjectType "ForeignGroup" -Scope  
"/subscriptions/<SUB_ID>" -Debug > newRoleAssignment.log
```

3.5. Notes

- A reseller relationship must be established before the foreign principal can be added.
- The user must have the ability to browse the Azure AD directory. This means the user must have at least the Directory Reader role.
- The user must be a local account in the tenant's Azure AD. The user cannot be a guest user or a foreign principal.
- The user must be an "Owner" of the subscription.
- AOBO must be added to each subscription directly. Using Management Groups is not supported.

3.6. Base Script

The script to add AOBO to a subscription is created by taking the base script and then adding the AdminAgent ID, role name and subscription id.

This script adds the AOBO permissions to one subscription. It must be run for each subscription in the tenant.

```
New-AzRoleAssignment -RoleDefinitionName "<ROLE_NAME>" -ObjectId  
"<ADMINAGT_OBJECT_ID>" -ObjectType "ForeignGroup" -Scope  
"/subscriptions/<SUB_ID>"
```

When replacing the placeholders, you must leave the quotation marks and remove the < and > symbols.

<ROLE_NAME> - replace with a role name from Section 3.2. Capitalization and spacing apply.

<ADMINAGT_OBJECT_ID> - replace with the appropriate object ID from Section 3.1

<SUB_ID> - replace with the subscription id from the end user tenant

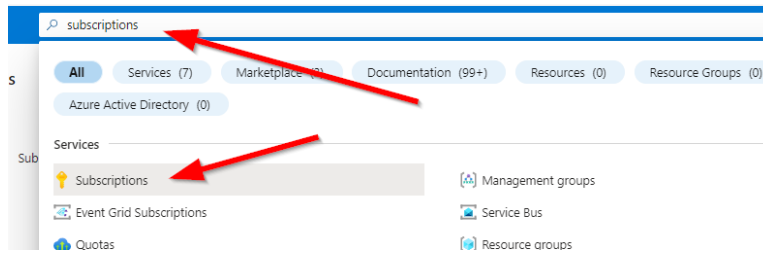
3.7. Finding Subscription IDs

- Log into the [Azure portal](#) and select the tenant containing the subscription
- Search for Subscriptions from the search bar then click on Subscriptions

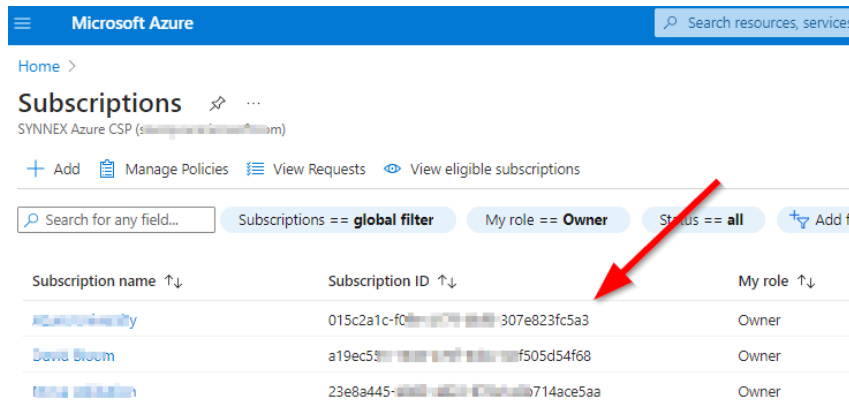
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- The subscription IDs will be shown in the list. They may be copied from this list to complete the AOBO script.



3.8. Establishing the Administration on Behalf Of (AOBO) aka Foreign Principal

IMPORTANT: TD Synnex must have a reseller relationship before you can add our foreign principal.

IMPORTANT: A user with Owner RBAC right to the subscription will need to run the script.

- This script will need to be run for each subscription under TD SYNEX.**
- The script may be run using the cloud shell:

- Log into the Azure portal for the end user tenant.
- Create the AOBO script per Section 3.1-3.4 above.
- Start Cloud Shell by clicking on the icon in the portal.

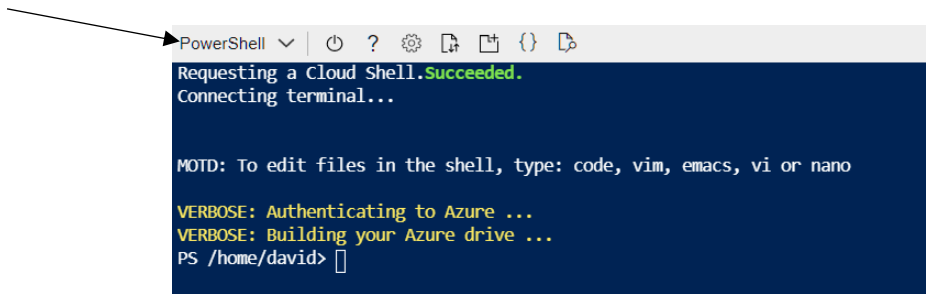


- When you start Power Shell for the first time, you will need to set up a Storage account for it to use.
- Confirm that you are in the correct shell: PowerShell or Bash. If you want to use the Power Shell command, you need to be in PowerShell and if you want to use the Azure CLI command, you can switch to Bash or execute "az" in PowerShell first.

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```
PowerShell
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

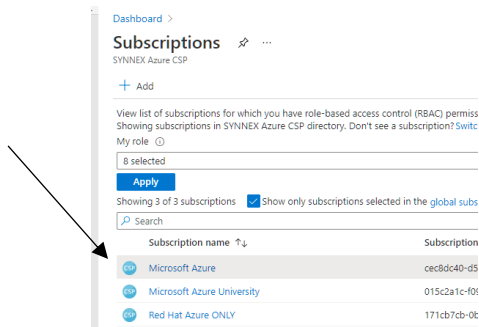
MOTD: To edit files in the shell, type: code, vim, emacs, vi or nano

VERBOSE: Authenticating to Azure ...
VERBOSE: Building your Azure drive ...
PS /home/david>
```

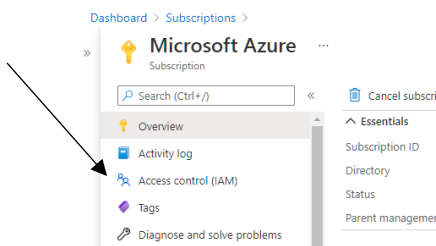
- Paste the command from above into the shell. You will need to right click in the shell then click “Paste” to paste the command.
- Press enter to submit the command.
- The shell will process the command then provide a success or failure message. If the command failed, you may contact the TD SYNEX DSE team for assistance.
- Close the shell window by clicking the X in the upper right corner of the shell window.

3.9. Removing AOBO (when required)

- A user with Owner RBAC right to the subscription will need to complete the following steps for each subscription.
- Navigate to the Subscriptions blade in the [Azure Portal](#)
- Click on the Subscription that contains the foreign principal to remove



- Click on Access Control (IAM)

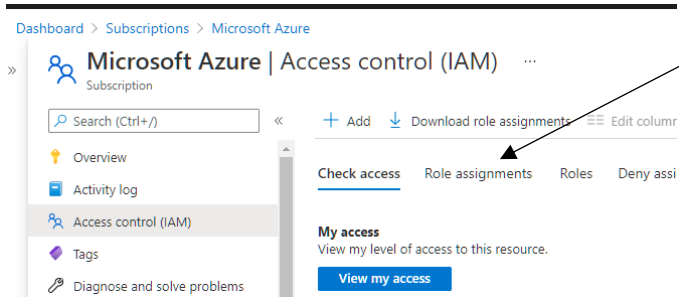


- Click on Role assignments

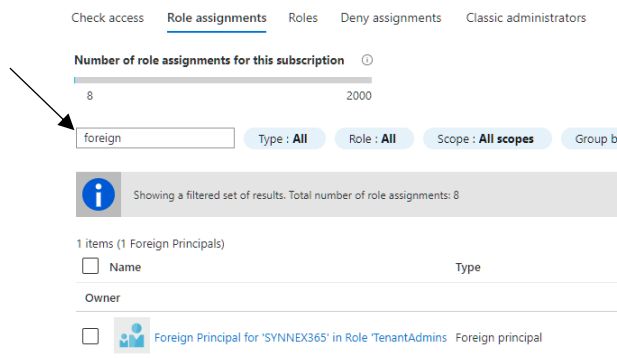
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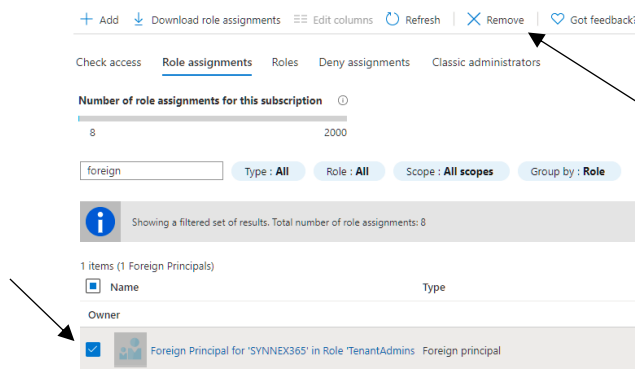


- In the search box, enter foreign to filter on just foreign principals



- There may be multiple foreign principals listed. Check the box by the one to be removed, then click Remove.

CAUTION: Do not remove the current CSP Provider. Removing the current provider may affect support and cost.



4. Associated Documents

- [Azure plan - Manage subscriptions & resources - Partner Center | Microsoft Learn](#)
- [Assign Azure roles using Azure PowerShell - Azure RBAC | Microsoft Learn](#)
- [New-AzRoleAssignment \(Az.Resources\) | Microsoft Learn](#)
- [Using the Azure Cloud Shell window | Microsoft Learn](#)
- [Quickstart for Azure Cloud Shell | Microsoft Learn](#)

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- [Partner Earned Credit Overview](#)
- [Partner Earned Credit Calculation](#)
- [Partner Earned Credit - Permissions](#)

5. Document Approval

This document has been approved by:

- Manager, Cloud Sales Engineering

6. Document Revision History

Number	Effective Date	Revision History	Author
1.0	4/7/2023	Initial Draft	David Bloom
1.1	7/12/2023	Corrected spelling and reseller relationship requirement	David Bloom
2.0	7/25/2023	Added error messages	David Bloom
2.1	8/21/2023	Added Reservation Purchaser	David Bloom
2.2	9/26/2023	Added note about Management Groups	David Bloom
3.0	7/3/2024	Added explanation of AOBO	David Bloom
3.1	8/21/2024	Added Appendix for CloudShell	David Bloom

Appendix A – Full Owner Scripts

IMPORTANT: Copy the script to Notepad and check for odd characters. Acrobat often replaces some characters which can cause the script to fail.

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Synnex (synnex365.com)

```
New-AzRoleAssignment -RoleDefinitionName "Owner" -ObjectId "b13ae2dd-2d47-4211-89de-5142f8368588" -ObjectType "ForeignGroup" -Scope
"/subscriptions/<subscription_id>"
```

TechData SIE

```
New-AzRoleAssignment -RoleDefinitionName "Owner" -ObjectId "aa7face0-42ae-47f1-a752-776945feb7fc" -ObjectType "ForeignGroup" -Scope
"/subscriptions/<subscription_id>"
```

TechData SCM

```
New-AzRoleAssignment -RoleDefinitionName "Owner" -ObjectId "9967c88e-30d5-476f-ab06-0b849ba52bde" -ObjectType "ForeignGroup" -Scope
"/subscriptions/<subscription_id>"
```

CANADA

Synnex

```
New-AzRoleAssignment -RoleDefinitionName "Owner" -ObjectId "acd31259-050b-4f6d-a1ba-0fe854bb0181" -ObjectType "ForeignGroup" -Scope
"/subscriptions/<subscription_id>"
```

TechData SCM

```
New-AzRoleAssignment -RoleDefinitionName "Owner" -ObjectId "28965048-c414-4fdb-b297-6e1c0d373331" -ObjectType "ForeignGroup" -Scope
"/subscriptions/<subscription_id>"
```

IMPORTANT: Replace <subscription id> with the customer's subscription ID

Appendix B – All Subscriptions in Tenant (PowerShell)

The following script will apply the requested role to all subscriptions in the tenant. Use this script when using PowerShell.

Note: If the role already exists for the adminAgentId, a “Conflict” error will be generated. This is expected and will not harm the system.

Note: You will get a “Forbidden” error for any subscription that you are not an owner. This is expected.

Note: If you are using this script in CloudShell, you will not use the Connect-AzAccount line.

```
$tenantid = ""
$adminAgentid = ""
$requestedRole = ""

$azAccount = (Connect-AzAccount -TenantId $tenantId -Force).Context.Tenant.Id

$subIds = (Get-AzSubscription -TenantId $tenantId).SubscriptionId

foreach ($subId in $subIds)
{
    New-AzRoleAssignment -RoleDefinitionName $requestedRole -ObjectId
    $adminAgentId -ObjectType "ForeignGroup" -Scope "/subscriptions/$subId"
}
```

In the first three lines of the script, add the appropriate information between the quotation marks.

\$tenantId – add the tenant id of the end user tenant

Note: The tenant Id may be found on the Overview tab in the Azure AD blade in Azure portal

\$adminAgentId – add the appropriate AdminAgent object id from Section 3.1

\$requestedRole – add the appropriate RBAC role name from Section 3.2

Appendix C – All Subscriptions in Tenant (CloudShell)

The following script will apply the requested role to all subscriptions in the tenant. Use this script when using CloudShell.

Note: If the role already exists for the adminAgentId, a “Conflict” error will be generated. This is expected and will not harm the system.

Note: You will get a “Forbidden” error for any subscription that you are not an owner. This is expected.

Note: If you are using this script in CloudShell, you will not use the Connect-AzAccount line.

```
$adminAgentid = ""
$requestedRole = ""

$subIds = (Get-AzSubscription).SubscriptionId

foreach ($subId in $subIds)
{
    New-AzRoleAssignment -RoleDefinitionName $requestedRole -ObjectId
    $adminAgentId -ObjectType "ForeignGroup" -Scope "/subscriptions/$subId"
}
```

In the first two lines of the script, add the appropriate information between the quotation marks.

\$adminAgentid – add the appropriate AdminAgent object id from Section 3.1

\$requestedRole – add the appropriate RBAC role name from Section 3.2

Appendix D – Limited Permissions - All Subscriptions in Tenant (CloudShell)

The following script will apply the requested roles to all subscriptions in the tenant. Use this script when using CloudShell.

Note: If the role already exists for the adminAgentId, a “Conflict” error will be generated. This is expected and will not harm the system.

Note: You will get a “Forbidden” error for any subscription that you are not an owner. This is expected.

Note: If you are using this script in CloudShell, you will not use the Connect-AzAccount line.

```
$adminAgentId = ""
$requestedRoles = "Support Request Contributor","Quota Request Operator",
"Reservations Administrator"

$subIds = (Get-AzSubscription).SubscriptionId

foreach ($subId in $subIds)
{
    Foreach($role in $requestedRoles)
    {
        New-AzRoleAssignment -RoleDefinitionName $role -ObjectId $adminAgentId -
        ObjectType "ForeignGroup" -Scope "/subscriptions/$subId"
    }
}
```

In the first line of the script, add the appropriate information between the quotation marks.

\$adminAgentId – add the appropriate AdminAgent object id from Section 3.1